

COUNT TO 15, THEN ANSWER ANY ESTIMATION QUESTION

How Many People Are in Times Square Right Now?

Dennis • Sr PM at Microsoft • @dennisthinks_

1 Buy time, then count to 15

Tell the interviewer you're taking a moment, then breathe and count to 15 in your head. Do it even if you already know how you'd start. Use the count to decide what you'd ask first.

EXAMPLE

"Give me a few seconds to think this through."

When you hit 15, don't start solving. Ask your question.

2 Ask clarifying questions

For an estimation, ask one or two. For a design question, three to five. Pick the ones that change your answer the most. "In Times Square" could mean everyone standing there right now or everyone who walks through today. In my math, those numbers are about 40 times apart.

EXAMPLE

"Do you mean people standing in Times Square right now, or everyone who walks through today? And what time of day?"

Repeat their answer back before you move on: "Got it, people standing in Times Square at 7pm on a normal night."

3 Break it apart on paper

Write the question at the top of the page and split it into layers before you do any math. Start with the biggest number you know and work down to the one they asked for. I'll assume most of the crowd is tourists, so tourists go at the top.

EXAMPLE

```
People in Times Square at 7pm
└ Tourists in NYC today: _____
  └ Share who walk through Times Square: ___%
    └ Plus New Yorkers walking through: _____
      └ Share of the day in the 7pm hour: ___%
        └ Minutes the average person stays: ___
```

Leave the blanks empty for now. Turn the page toward the interviewer so they can follow along.

4 Work it top down

Fill in the tree one layer at a time, out loud, and give a reason for every percentage. Round to numbers you can multiply in your head.

EXAMPLE

1. I'll guess NYC gets about 60 million tourists a year, and a trip lasts about 3 days.
 $60M \times 3 \text{ days} \div 365 \approx 500,000 \text{ tourists in the city today}$
2. Say half of them walk through Times Square today. It's on almost every first trip.
 $500,000 \times 50\% = 250,000$
3. Add New Yorkers. The subway station under it is one of the busiest in the city, and offices and theaters surround the square. Say 50,000.
 $250,000 + 50,000 = 300,000 \text{ walk through today}$
4. 7 to 8pm brings the dinner and Broadway crowds, so say 10% of the day comes through then.
 $300,000 \times 10\% = 30,000 \text{ in the 7pm hour}$
5. The average person stays about 15 minutes. Photos, the red steps, then they move on.
 $30,000 \times 15 \div 60 = 7,500 \text{ people right now}$

If they push back on a percentage, take their number and keep going.

5 Check your number with the interviewer

You can't look anything up in an interview, so the interviewer is your fact check. Say your number and ask if it's in the right range. They'll tell you it's about right, too high, or too low.

EXAMPLE

You: "That puts me at about 7,500 people at 7pm. Does that feel like the right range?"

Interviewer: "That's a little high."

You: "The number I'm least sure of is the 15 minutes. Most people walk through without stopping. At 10 minutes, it's $30,000 \times 10 \div 60$, about 5,000 people."

Interviewer: "That's closer."

If they say it's right, close with your number. If they say higher or lower, don't start over. Change the one assumption you're least sure of, out loud, and rerun the math from that layer down.

Practice on this one

Set a 5-minute timer, have a friend play the interviewer, and record it. Run all five steps out loud, count included. Play it back and flag any number you said without a reason.

☐ How many people are at the Atlanta airport right now?

Score every answer

Use this checklist after each practice run:

- ☐ Did I say "give me a few seconds" and count to 15?
- ☐ Did I ask one or two clarifying questions and repeat the scope back?
- ☐ Did I sketch the layers on paper before any math?
- ☐ Did I give a reason for every percentage?
- ☐ Did I say each line of math out loud?
- ☐ Did I ask if my number was in range and change one assumption at a time?